

Latest in Post-Quantum Cryptography Migration -- diff report

Generated: 2026-08-31T21:50:57.271240 | **Period:** since 2026-08-31 | **Leaves with news:** 3 of 99

Fresh developments: 6 (as_of within 14 days of the window) | **Superseded/conflicting:** 0 | **Background captured:** 8 (older material, first time in the knowledge base)

Reconfirmations in this period (no new information, not shown): 1.

Summary

Leaf	Fresh	Superseded	Conflicting	Background
Attestation protocols & silicon RoT	3	0	0	7
Platform code signing	3	0	0	0
Email security	0	0	0	1

Firmware & root of trust

Attestation protocols & silicon RoT (attestation-protocols)

New developments (3)

Caliptra

- First Caliptra 2.1 Subsystem silicon has successfully powered on as of August 19, 2026. ([Caliptra Blog](#), as of 2026-08-19, confidence: medium, source: grok)
- *New silicon-bring-up milestone for Caliptra 2.1 not previously recorded.*

OpenTitan

- OpenTitan introduces new top-level design 'Earl Grey 2' with enhanced hardened PQC support (ML-KEM, ML-DSA), CHERI memory safety, and RRAM, with initial implementation targeted for 2026 Q1-Q3. ([OpenTitan Roadmap](#), as of 2026-08-25, confidence: high, source: grok)
- *New OpenTitan top-level design (Earl Grey 2) not previously present in the knowledge base.*
- OpenTitan project announces new 'Peppermint' integrated root-of-trust top-level optimized for area-constrained SoCs (wearables, IoT), incorporating PQC support, with design in H2 2026 and completion in H1 2027. ([OpenTitan project kicks off the Peppermint integrated root-of-trust for area-constrained SoCs](#), as of 2026-08-25, confidence: medium, source: grok)
- *New OpenTitan top-level design (Peppermint) not previously present in the knowledge base.*

Newly captured background -- older material, first time in the knowledge base (7)

Caliptra

- Caliptra v2.x FIPS certification of cryptographic algorithms completed as of July 17, 2026. ([Caliptra Blog](#), as of 2026-07-17, confidence: medium, source: grok)
- *New certification-completion milestone for Caliptra v2.x not previously recorded.*

OTBN hardware extensions for ML-DSA and ML-KEM — approved, landing in 2026

- OpenTitan partners have approved hardware extensions to the OpenTitan Big Number accelerator (OTBN) to enable ML-DSA (FIPS 204) and ML-KEM (FIPS 203) implementations: expanded data memory (4 KiB to 32 KiB), a direct interface to the side-channel-hardened KMAC block, a 32-bit SIMD ISA extension for polynomial arithmetic/NTT acceleration, and automated hardware mask conversion between masking schemes. ([lowrisc.org — opentitans transition to side channel resilient post quantum cryptography](#), as of 2026-02-01, confidence: medium, source: ingest)
- *First mention of OTBN hardware extensions for ML-DSA/ML-KEM in the knowledge base.*
- The mask-conversion hardware exists because side-channel hardening otherwise dominates cost: lowRISC cites masking runtime overhead that could reach roughly 90% without it; the stated motivation for the whole extension set is side-channel-resilient PQC signing (not just verification) on the root of trust. ([lowrisc.org — opentitans transition to side channel resilient post quantum cryptography](#), as of 2026-02-01, confidence: medium, source: ingest)
- *New engineering rationale/detail not previously recorded, distinct from the general extension description.*
- The hardware changes are planned to complete in 2026, with work already underway in the OpenTitan GitHub master branch as of February 2026; contributors named include Google, Nuvoton, and academic teams including ETH Zurich's Integrated Systems Laboratory. ([lowrisc.org — opentitans transition to side channel resilient post quantum cryptography](#), as of 2026-02-01, confidence: medium, source: ingest)
- *New timeline/contributor detail not previously present.*

OpenTitan Earlgrey — shipped SLH-DSA secure boot

- OpenTitan's first production silicon, the "Earlgrey" chip, supports hardware-accelerated SPHINCS+ signature verification for secure boot, and the implementation is compliant with FIPS 205 SLH-DSA as standardized by NIST in 2024. ([lowrisc.org — opentitans transition to side channel resilient post quantum cryptography](#), as of 2026-02-01, confidence: medium, source: ingest)
- *No prior claim in the base or deltas covers OpenTitan/Earlgrey's SLH-DSA secure-boot verification capability.*
- Mass production of Earlgrey by Taiwanese manufacturer Nuvoton began in February 2025 per Google's announcement, with the chips slated for use as the hardware root of trust in new Chromebooks and Google's data-center servers, meaning the SLH-DSA-verified boot path is shipping in consumer and infrastructure hardware, not sitting in a reference design. ([heise.de — Open and quantum safe RISC V security chip OpenTitan for Chromebooks 10274833](#), as of 2025-02-28, confidence: medium, source: ingest)
- *New deployment/production detail not previously present in the knowledge base.*

ZeroRISC commercial stack

- ZeroRISC, with the Max Planck Institute for Security and Privacy and Academia Sinica, announced on 2026-03-09 the open-source release of a cryptographic hardware/software stack for classical and post-quantum operations — the Cryptolib embedded library paired with a programmable Asymmetric Cryptography Coprocessor (ACC) with hardware-accelerated ML-KEM, ML-DSA, and SLH-DSA, reporting 6–9x speedups and 36–75% higher max operating frequency at near-zero area cost. ([businesswire.com — ZeroRISC and Leading Research Institutions Deliver Production Grade Post Quantum Cryptography for Open Silicon](#), as of 2026-03-09, confidence: medium, source: ingest)
- *No prior mention of ZeroRISC's PQC hardware/software stack in the knowledge base.*

Software supply chain

Platform code signing (platform-code-signing)

New developments (3)

C2PA content credentials — PQ signing planned, not shipped

- The C2PA specification's own explainer (current specification line 2.4) states: "Future support for ML-DSA algorithms (ML-DSA-44, ML-DSA-65, and ML-DSA-87) with approved parameters, as defined in NIST's FIPS 204, is planned for the C2PA standard." ([spec.c2pa.org — Explainer](https://spec.c2pa.org/Explainer), as of 2026-08-31, confidence: medium, source: ingest)
- *No C2PA content-credentials coverage exists in the base snapshot; this establishes C2PA's PQC plans as a new subtopic.*
- The stated gate for adoption is library maturity, not standardization: ML-DSA will be added to the supported signature algorithm list "when stable ML-DSA support is broadly available in commonly used cryptographic libraries." No target date is given. ([spec.c2pa.org — Explainer](https://spec.c2pa.org/Explainer), as of 2026-08-31, confidence: medium, source: ingest)
- *New detail on C2PA's adoption criteria for PQC, not previously captured anywhere in the base.*
- Current C2PA signing rests on standard X.509-based credentials with SHA2-256-class hashing in a Merkle-tree-like manifest structure; no post-quantum signature algorithm is in the currently supported set. ([spec.c2pa.org — Explainer](https://spec.c2pa.org/Explainer), as of 2026-08-31, confidence: medium, source: ingest)
- *New baseline-state fact describing C2PA's current (non-PQC) cryptographic construction, absent from the existing knowledge base.*

Email & messaging

Email security (**email-security**)

Newly captured background -- older material, first time in the knowledge base (1)

DirectTrust / Direct Secure Messaging

- DirectTrust describes Direct Secure Messaging as a secure communication transport for sensitive information over the open internet that utilizes digital certificates and a public key infrastructure to encrypt message contents so only the intended recipient can decrypt them, forming the certificate-anchored trust fabric any future PQ transition for the ecosystem would run through. ([directtrust.org — direct secure messaging](https://directtrust.org/direct-secure-messaging), as of undated, confidence: low, source: ingest)
- *First claim in the base introducing DirectTrust/Direct Secure Messaging, a new subtopic area not previously covered.*